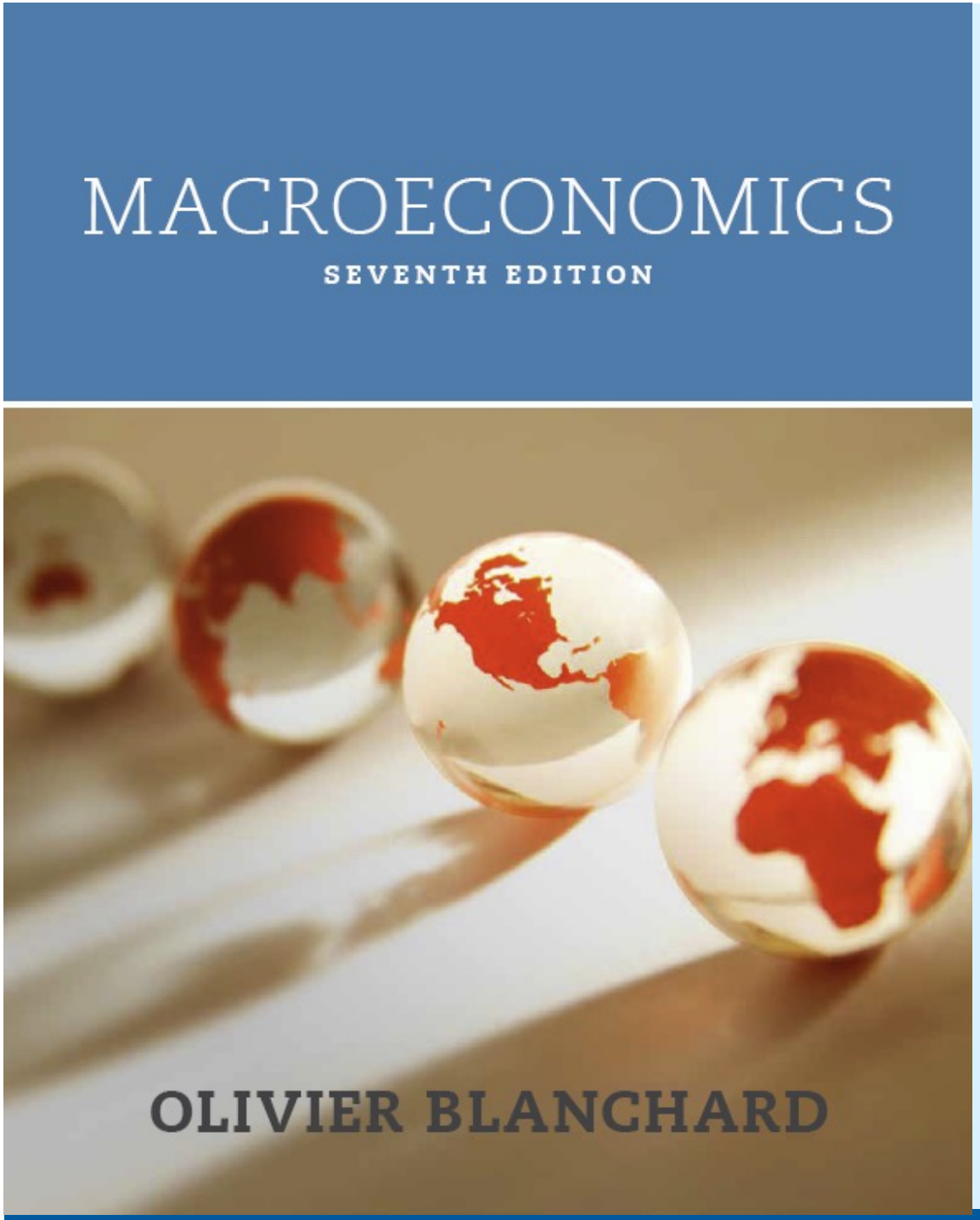
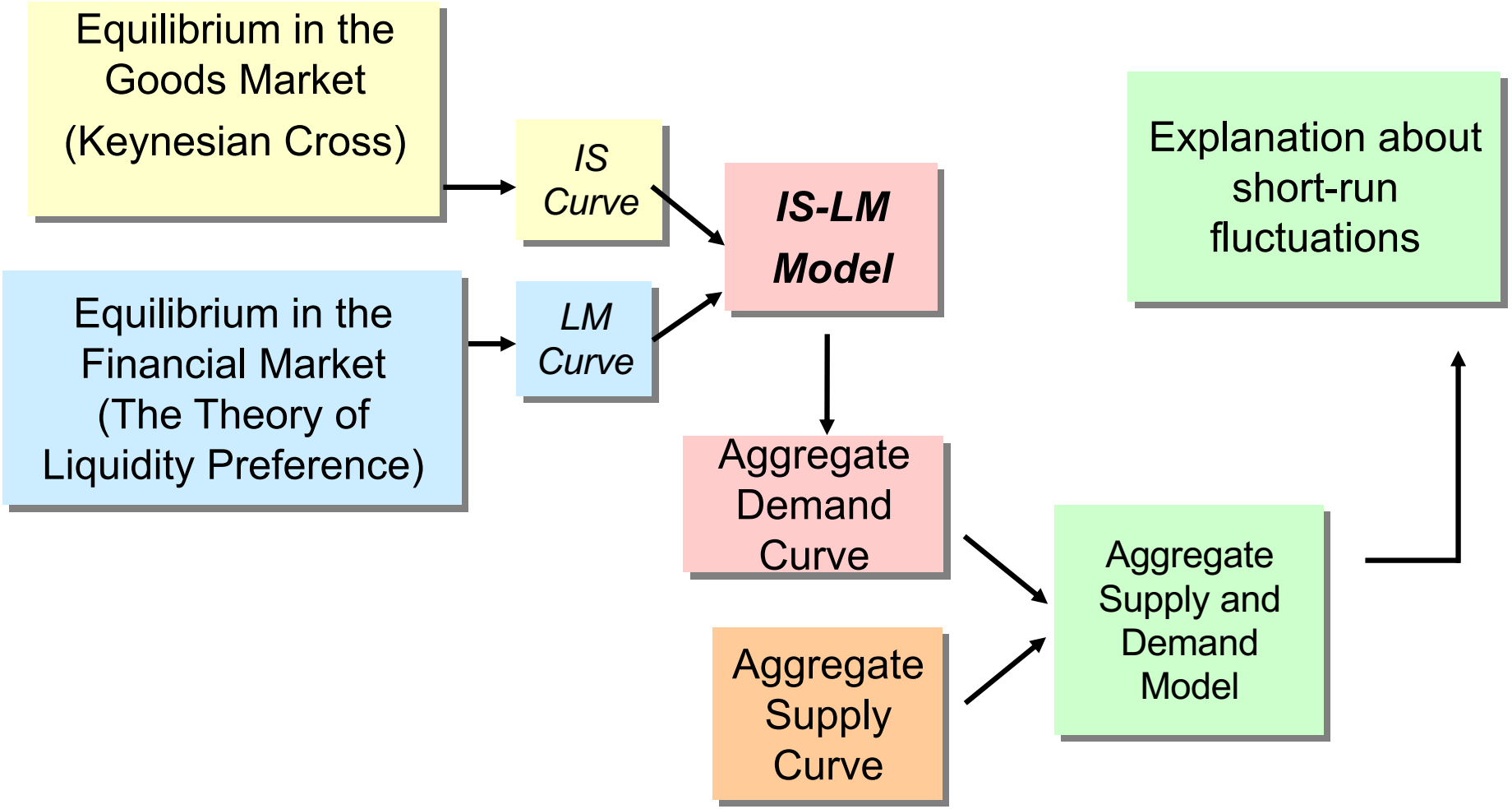


Goods and Financial Markets; The *IS-LM* Model

Chapter 5

ALWAYS LEARNING
PEARSON

General outline



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Chapter 5 Outline

Goods and Financial Markets: The *IS-LM* Model

5-1 The Goods Market and the *IS* Relation

5-2 Financial Markets and the *LM* Relation

5-3 Putting the *IS* and the *LM* Relations Together

5-4 Using a Policy Mix

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Goods and Financial Markets: The *IS-LM* Model

- We looked at the goods market in Chapter 3, and financial markets in Chapter 4.
- In this chapter, we look at goods and financial markets together, and understand how output and the interest rate are determined in the short run.
- John Hicks and Alvin Hansen called this framework the *IS-LM* model.

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5-1 The Goods Market and the *IS* Relation

- In the model developed in Chapter 3, investment was assumed to be constant for simplicity.
- In fact, investment depends on production Y (or sales) and the interest rate i.

$$I = I(Y,\ i) \qquad (5.1)$$
$$(+,-)$$

- So equilibrium in the goods market becomes:

$$Y = C(Y - T) + I(Y, i) + G \qquad (5.2)$$

which is the *IS* relation.

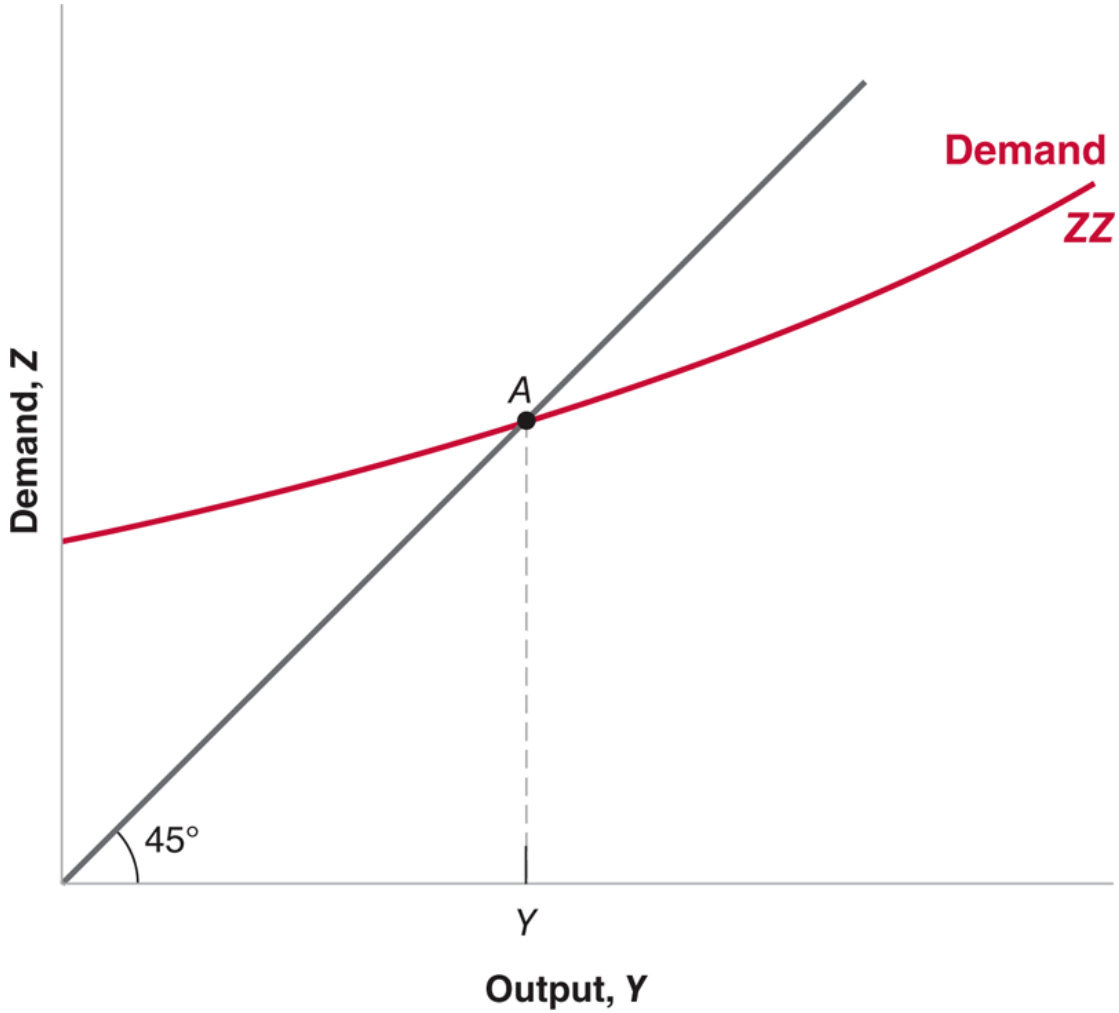
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5-1 The Goods Market and the *IS* Relation

Figure 5-1 Equilibrium in the Goods Market

The demand for goods is an increasing function of output.

Equilibrium requires that the demand for goods be equal to output.



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5-1 The Goods Market and the *IS* Relation

- ZZ is upward-sloping because, for a given value of the interest rate, an increase in output leads to an increase in the demand for goods through its effects on consumption and investment.
- ZZ is a curve rather than a line because we have not assumed that the consumption and investment relations in equations (5.2) are linear.
- ZZ is flatter than the 45-degree line because we have assumed that an increase in output leads to a less than one-for-one increase in demand.
- The intersection of ZZ and the 45-degree line (point *A*) is the equilibrium level of output.

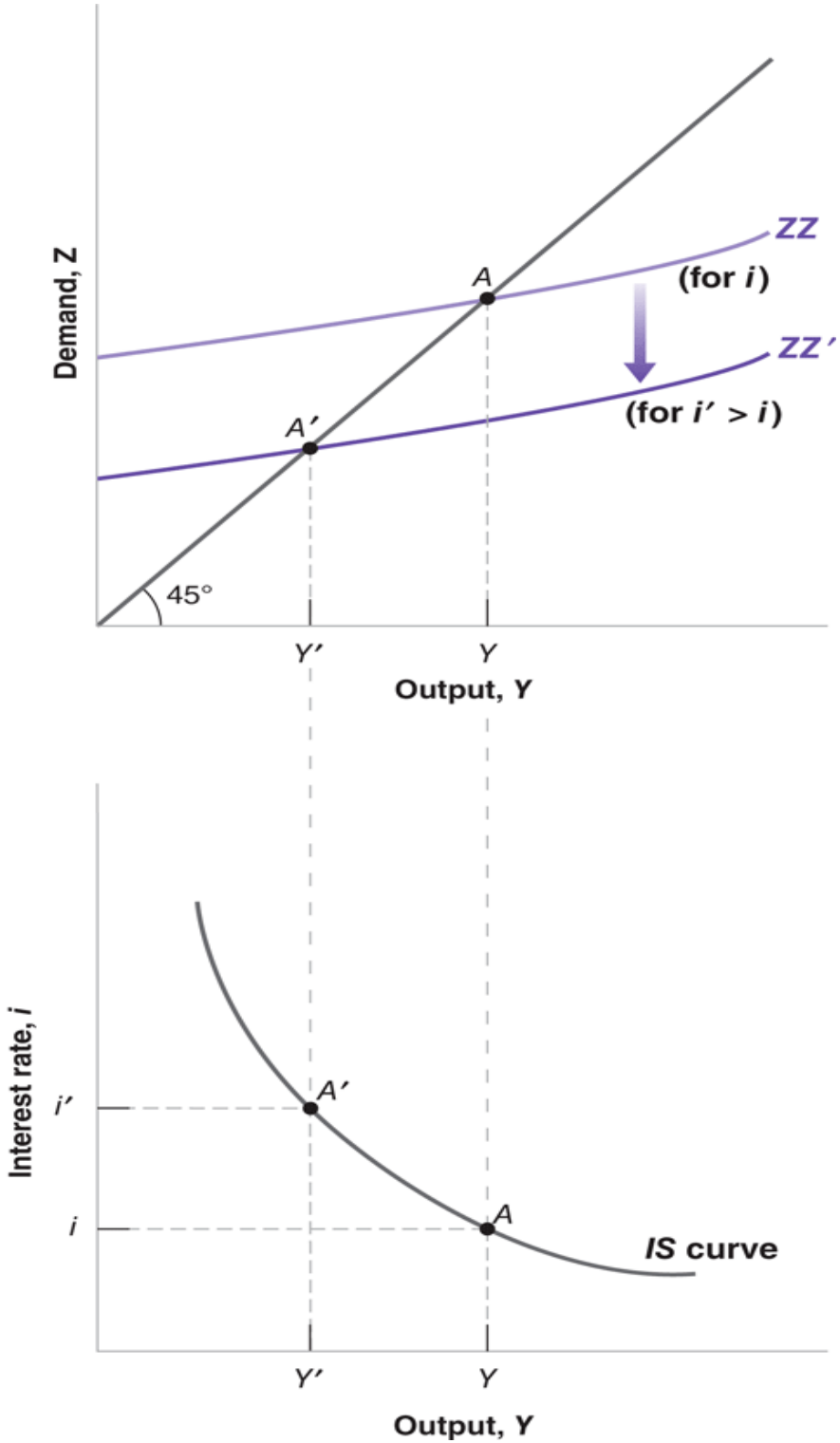
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5-1 The Goods Market and the *IS* Relation

(a) An **increase in the interest rate** decreases the demand for goods at any level of output, leading to a **decrease in the equilibrium level of output.**

(b) Equilibrium in the goods market implies that an increase in the interest rate leads to a decrease in output.

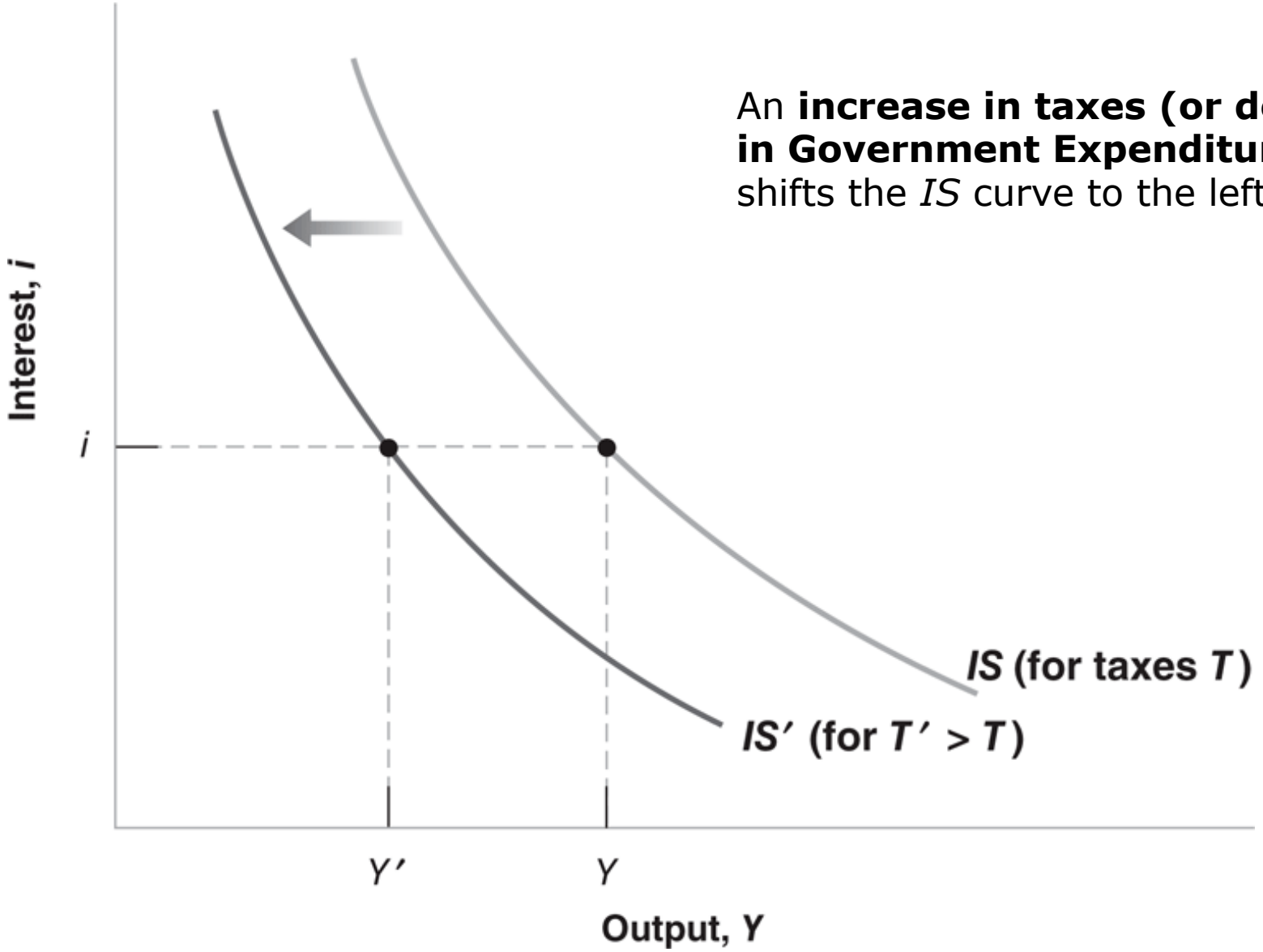
The ***IS* curve** is therefore downward sloping.



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5-1 The Goods Market and the *IS* Relation

Shifts of the *IS* Curve



An **increase in taxes (or decrease in Government Expenditure)** shifts the *IS* curve to the left.

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5-1 The Goods Market and the *IS* Relation

- Downward-sloping *IS* curve: **Equilibrium in the goods market** implies that an increase in the interest rate leads to a decrease in output.

- Shifting the *IS* curve: Changes in factors that decrease (increase) the demand for goods given the interest rate shift the *IS* curve to the left (right).

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5-2 Financial Markets and the *LM* Relation

- Recall Chapter 4: $M = \$Y\,L(i)$
- Divide both sides of the equation by the price level P:

$$\frac{M}{P} = Y\,L(i) \qquad (5.3)$$

  which is the *LM* relation.
- **In equilibrium, real money supply equals the real money demand**, which depends on real income Y, and the interest rate i.

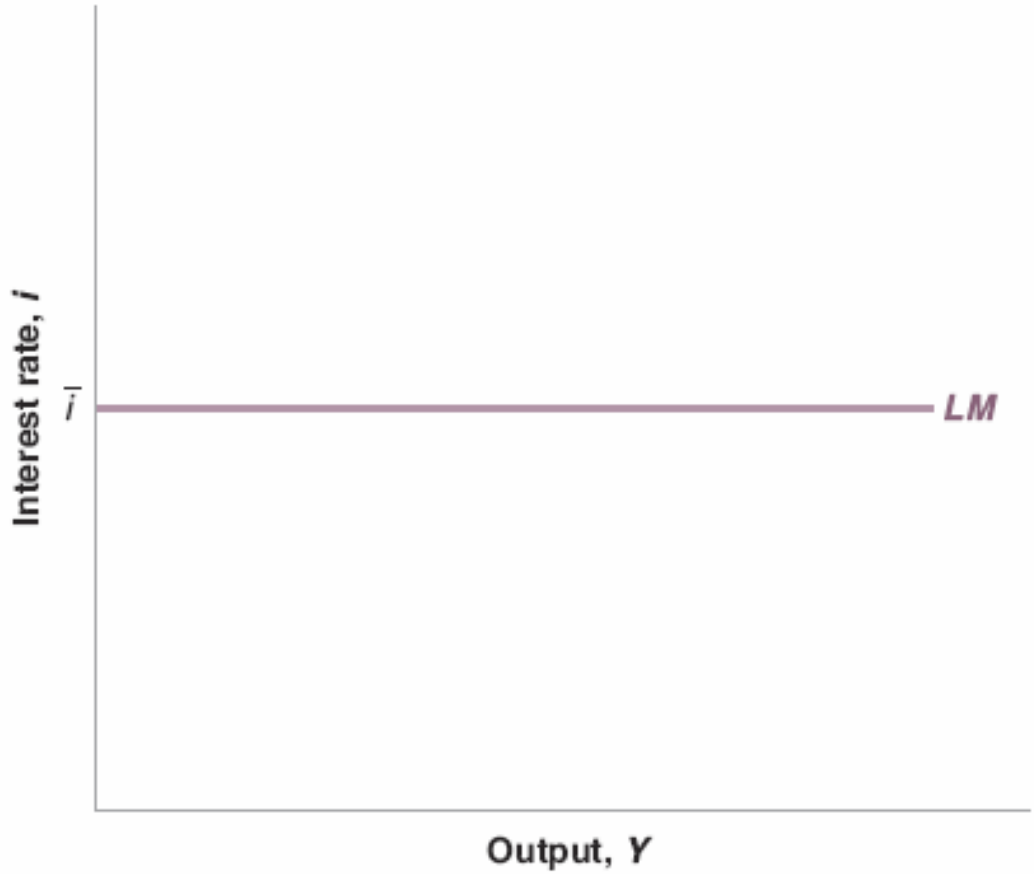
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5-2 Financial Markets and the *LM* Relation

Figure 5-4 The *LM* Curve

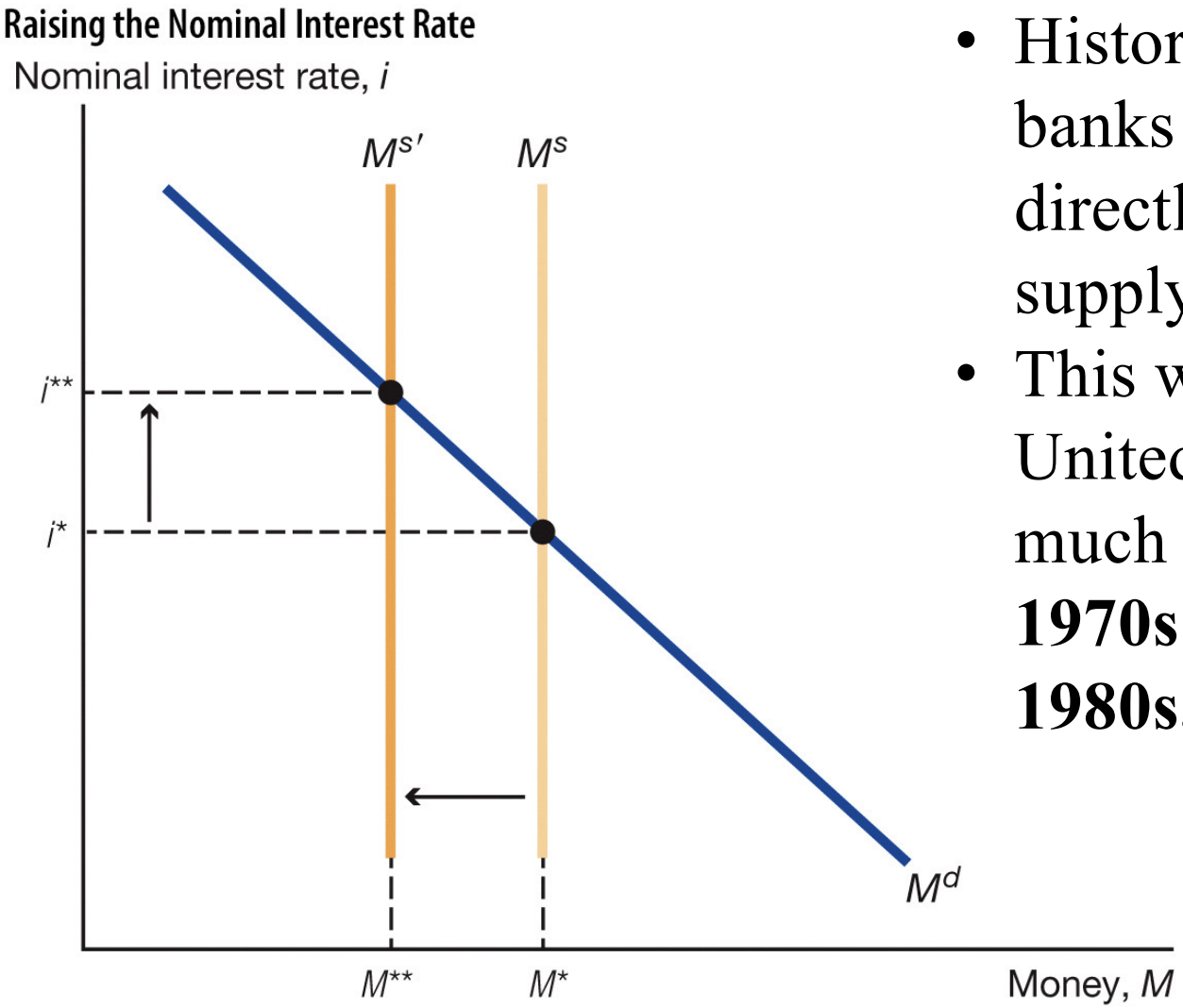
The central bank chooses the interest rate (and adjusts the money supply so as to achieve it).

Before we have seen that if Y changes, I adjusts so M/P remains fixed. Now, the Central Bank chooses i and adjusts M/P.



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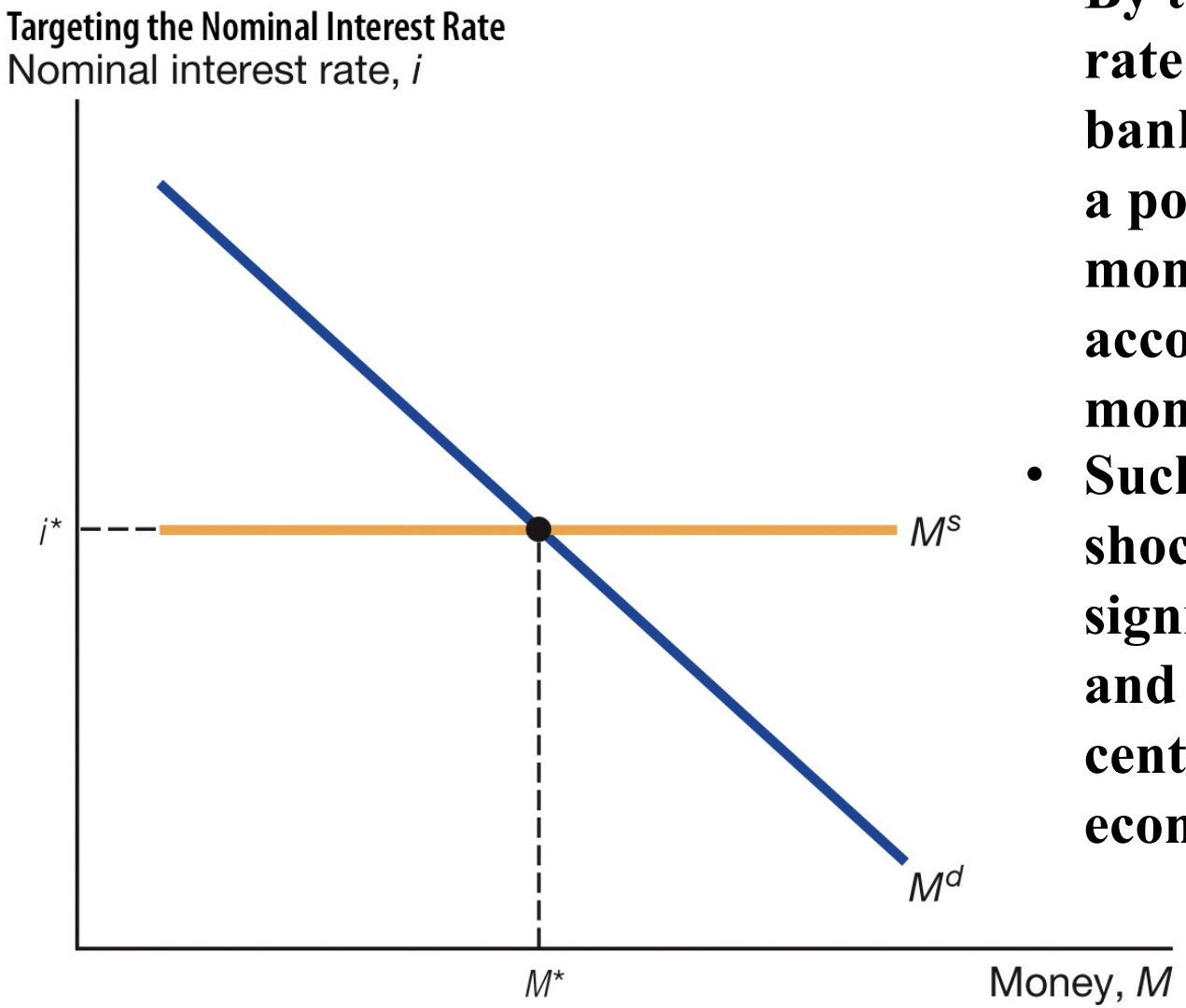
BEFORE: Raising the Nominal Interest Rate



- Historically, central banks did in fact focus directly on the money supply.
- This was true in the United States and much of Europe in the **1970s and early 1980s.**

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NOW: Targeting the Nominal Interest Rate



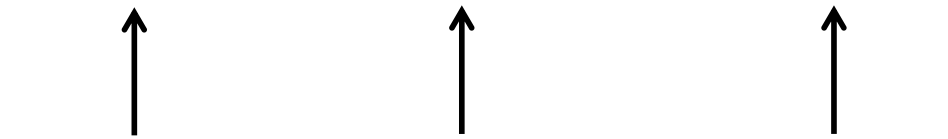
- By targeting the interest rate directly, the central bank automatically adopts a policy that will adjust the money supply to accommodate shocks to money demand.
- Such a policy prevents the shocks from having a significant effect on output and inflation and helps the central bank stabilize the economy.

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From Nominal to Real Interest Rates

- Fisher equation:

$$i_t = R_t + \pi_t$$



Nominal interest rate

Real interest rate

Inflation

$$R_t = i_t - \pi_t$$

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From Real to Nominal Interest Rates

- Sticky inflation assumption.
- In the short run, inflation
  - displays inertia, or stickiness
  - adjusts slowly over time
  - does not respond directly to monetary policy

- Central banks can set the real interest rate in the short run

$$\text{If} \qquad \pi = 0 \qquad \Longrightarrow \qquad R_t = i_t$$

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5-2 Financial Markets and the *LM* Relation

- *IS* relation: $Y = C(Y-T) + I(Y, i) + G$
- *LM* relation: $i = \bar{i}$
- The *IS* and *LM* relations together determine output.

- *Any point* on the downward sloping *IS* curve corresponds to equilibrium in the goods market.
- *Any point* on the horizontal *LM* curve corresponds to equilibrium in financial markets.
- *Only at* their intersection (*point A*) are both equilibrium relations satisfied.

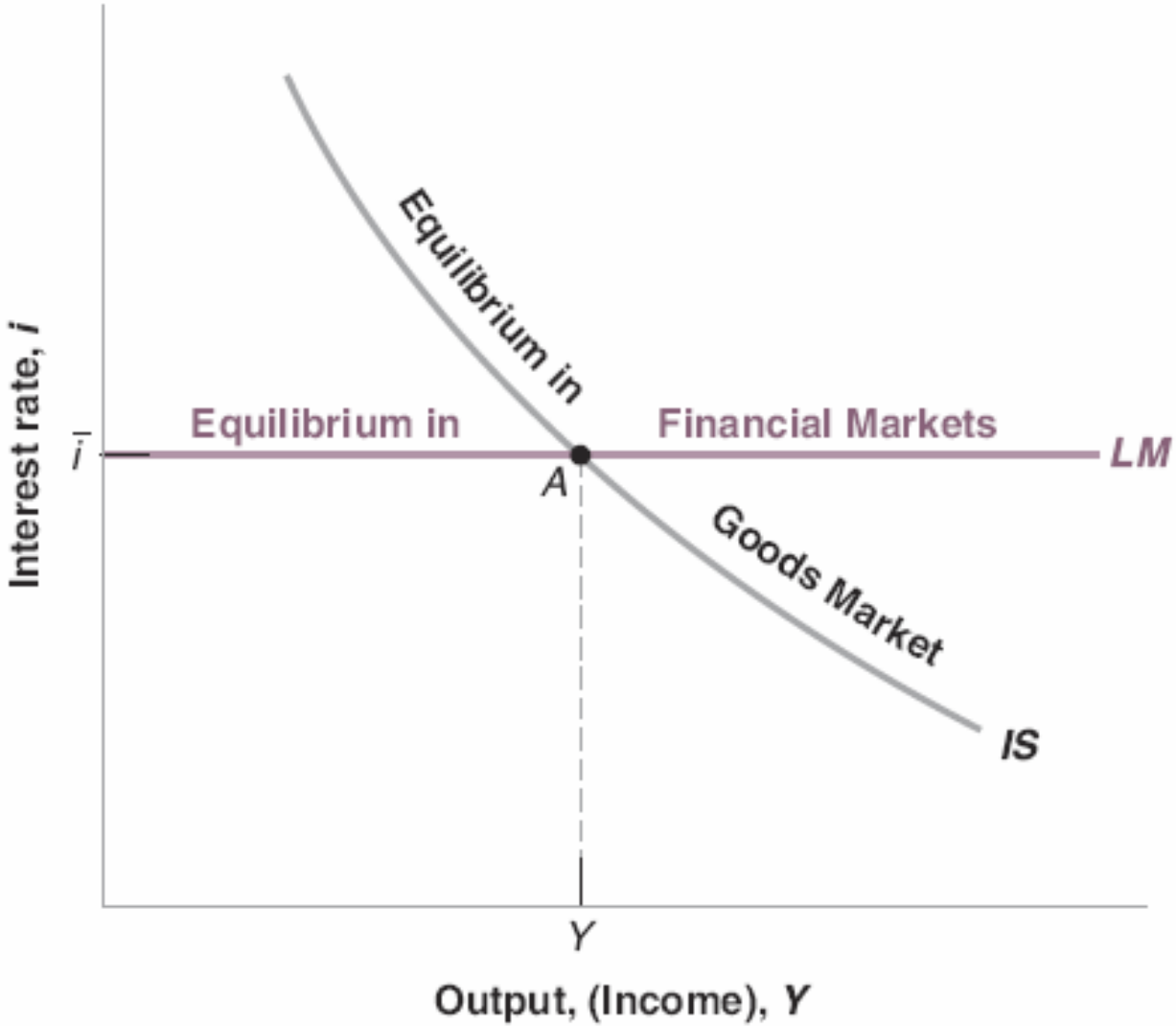
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5-3 Putting the *IS* and the *LM* Relations Together

- Equilibrium in the goods market implies that an increase in the interest rate leads to a decrease in output. This is represented by the *IS* curve.

- Equilibrium in financial markets is represented by the horizontal *LM* curve. Only at point *A*, which is on both curves, are both goods and financial markets in equilibrium.

The *IS–LM* Model



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5-3 Putting the *IS* and the *LM* Relations Together

What is the effect of Fiscal & Monetary Policy?

ANALYSIS OF SHORT-TERM
ECONOMIC POLICIES
TO STABILIZE THE ECONOMY

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5-3 Putting the *IS* and the *LM* Relations Together

- Fiscal Policy:

  Decrease in $G-T$ **fiscal contraction**
  fiscal consolidation

  Increase in $G-T$ **fiscal expansion**

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5-3 Putting the *IS* and the *LM* Relations Together

- Steps for analyzing the effects of changes in policy or exogenous variables:

  1. Does it shift the *IS* curve and/or the *LM* curve?

  2. What does this do to equilibrium output and the equilibrium interest rate?

  3. Describe the effects in words.

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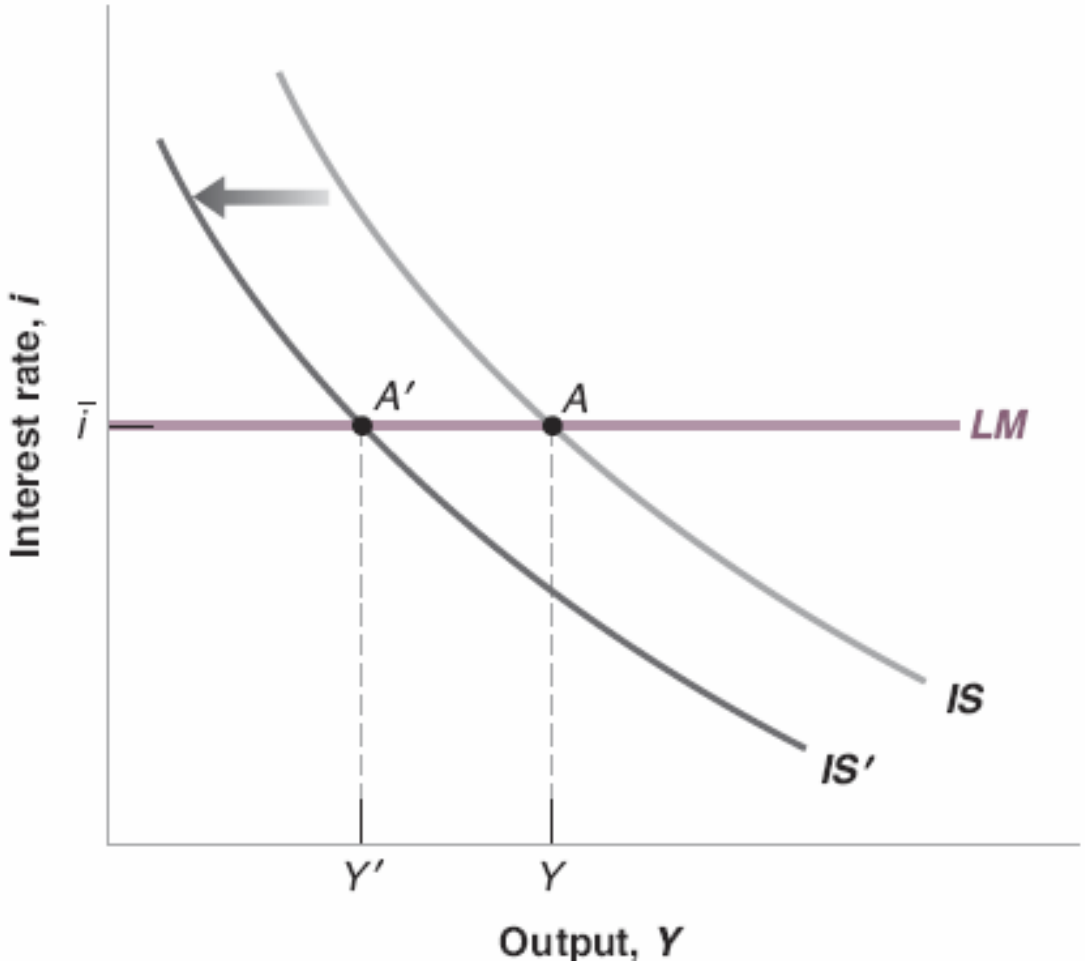
5-3 Putting the *IS* and the *LM* Relations Together

Figure 5-6 The Effects of an Increase in Taxes

An increase in taxes shifts the *IS* curve to the left.

This leads to a decrease in the equilibrium level of output.

Remember the Multiplier effect from the equilibrium in the Goods Market.



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5-3 Putting the *IS* and the *LM* Relations Together

- Monetary Policy:

Decrease in *i* increase in *M* **monetary expansion**

Increase in *i* decrease in *M* **monetary contraction**

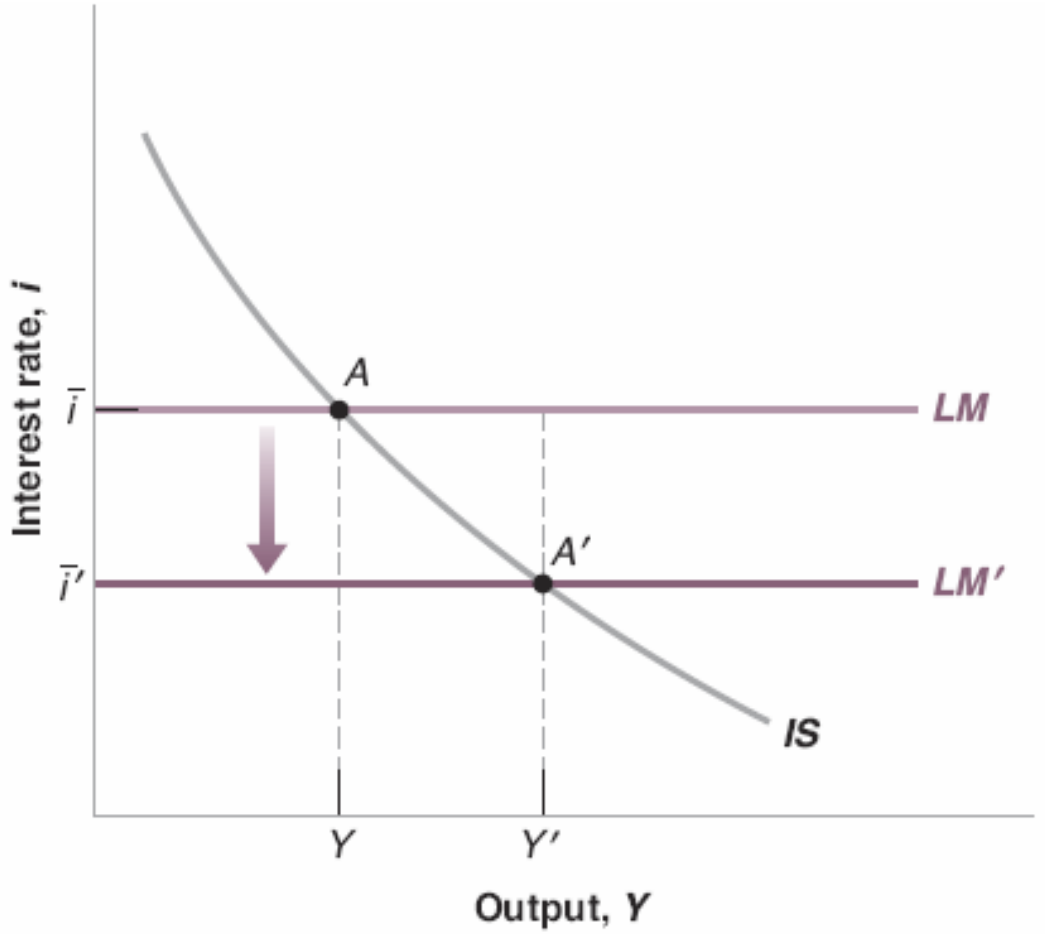
monetary tightening

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5-3 Putting the *IS* and the *LM* Relations Together

Figure 5-7 The Effects of a Decrease in the Interest Rate

A monetary expansion shifts the *LM* curve down, and leads to higher output.



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5-4 Using a Policy Mix

- **Monetary-fiscal policy mix** is the combination of monetary and fiscal policies.
- Suppose that the economy is in a recession and output is too low.
- Both fiscal and monetary policies can be used to increase output.

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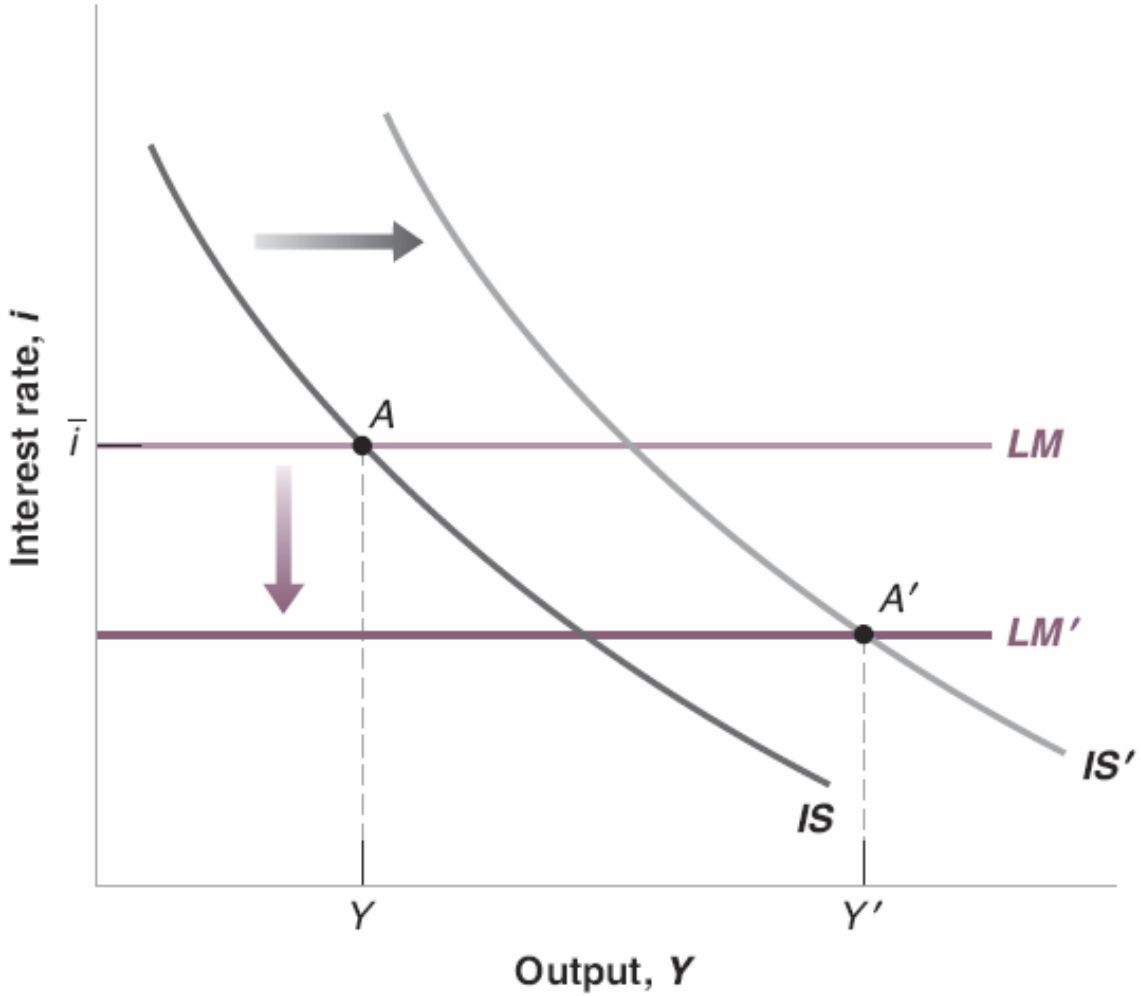
5-4 Using a Policy Mix

Figure 5-8 The Effects of a Combined Fiscal and Monetary Expansion

The fiscal expansion shifts the *IS* curve to the right.

A monetary expansion shifts the *LM* curve down.

Both lead to higher output.

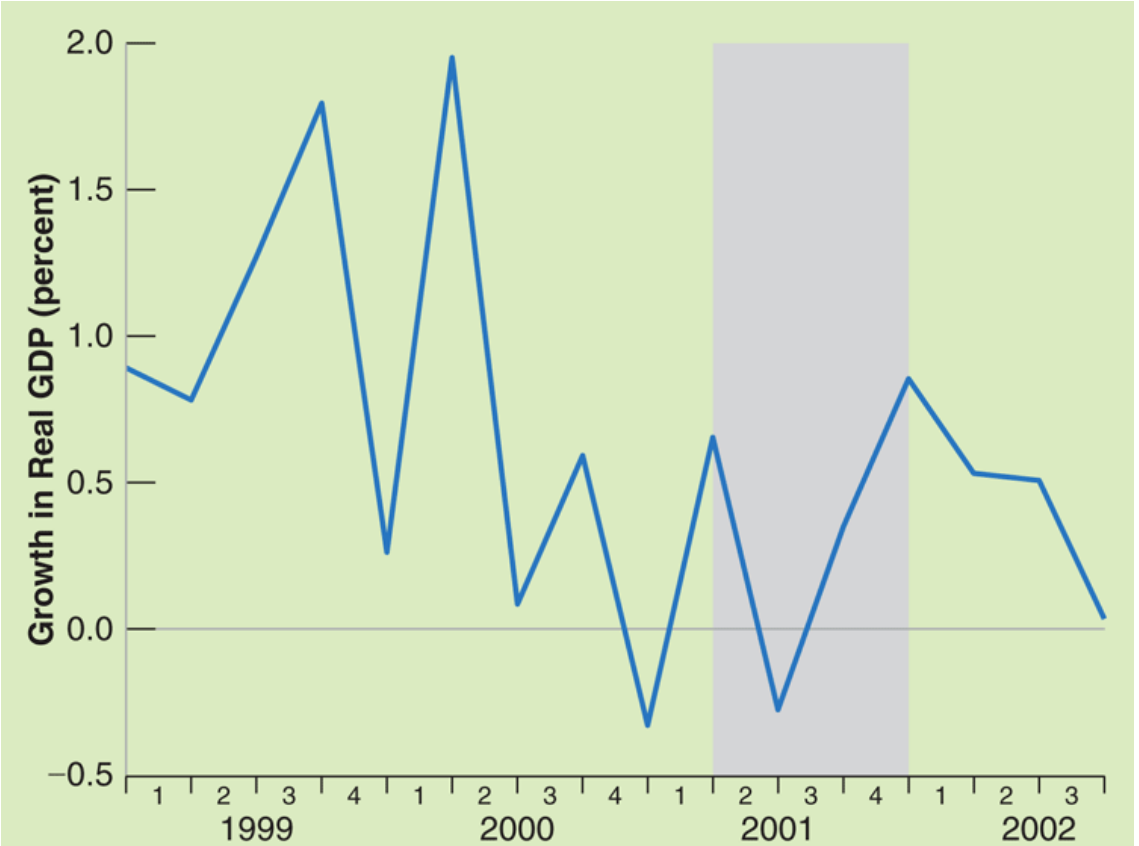


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Example: The U.S. Recession of 2001

Figure 1 The U.S. Growth Rate, 1999–1 to 2002–4

The National Bureau of Economic Research concluded that the U.S. economy was in a recession between March 2001 and December 2001, triggered by sharp declines in investment demand.



Source: Calculated using Series GDPC1, Federal Reserve Economic Data (FRED) http://research.stlouisfed.org/fred2/

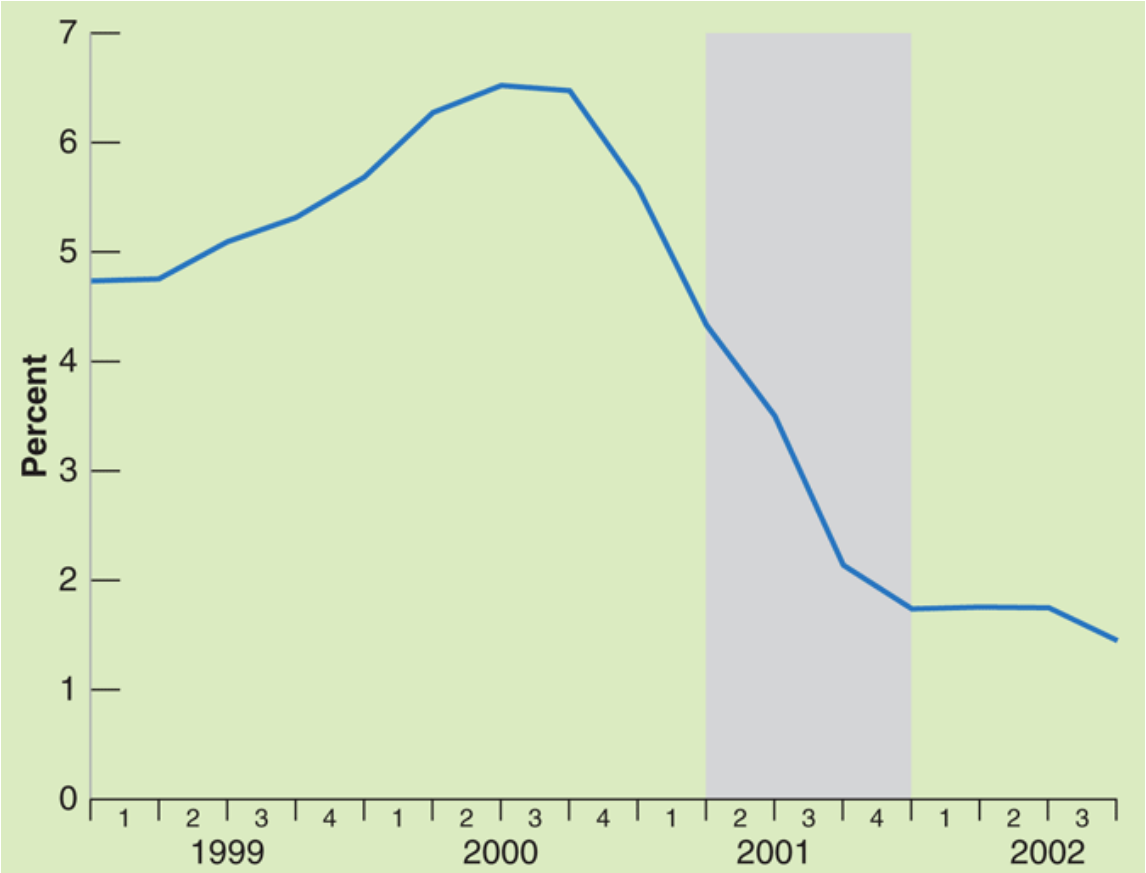
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Example: The U.S. Recession of 2001

Figure 2 The Federal Funds Rate, 1999–1 to 2002–4

The recession was met by strong macroeconomic policy response.

The Fed cut the federal funds rate from **6.5%** in January to **2%** at the end of 2001.

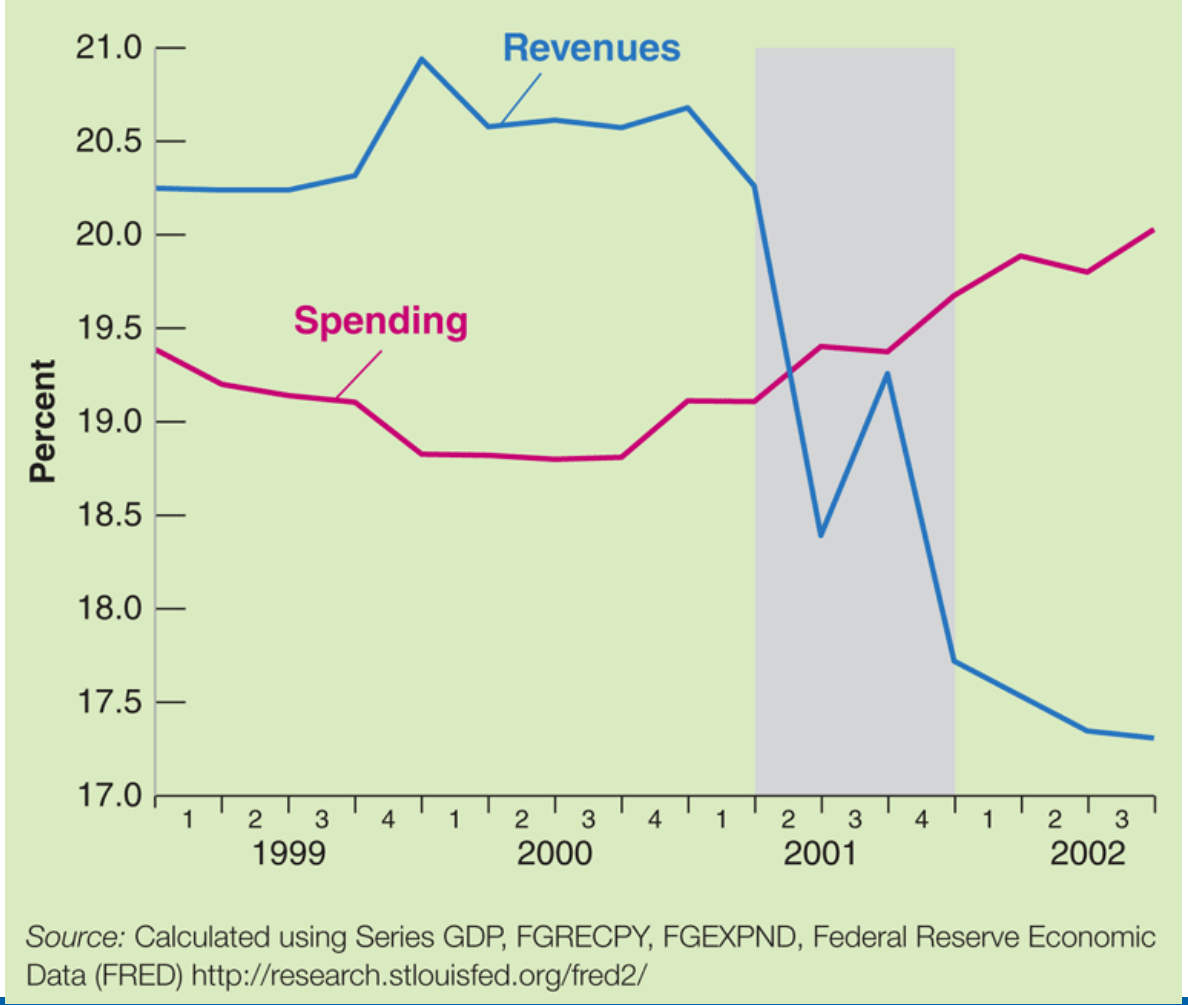


Source: Series FEDFUNDS, Federal Reserve Economic Data (FRED) http://research.stlouisfed.org/fred2/

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Example: The U.S. Recession of 2001

Figure 3 U.S. Federal Government Revenues and Spending (as Ratios to GDP), 1999–1 to 2002–4



Source: Calculated using Series GDP, FGRECPY, FGEXPND, Federal Reserve Economic Data (FRED) http://research.stlouisfed.org/fred2/

President George Bush also **cut taxes** in 2001 and 2002 budgets.

The events of September 11, 2001 also lead to an **increase in spending** on defense and homeland security.

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5-4 Using a Policy Mix

Figure 5-9 The Effects of a Combined Fiscal Consolidation and a Monetary Expansion

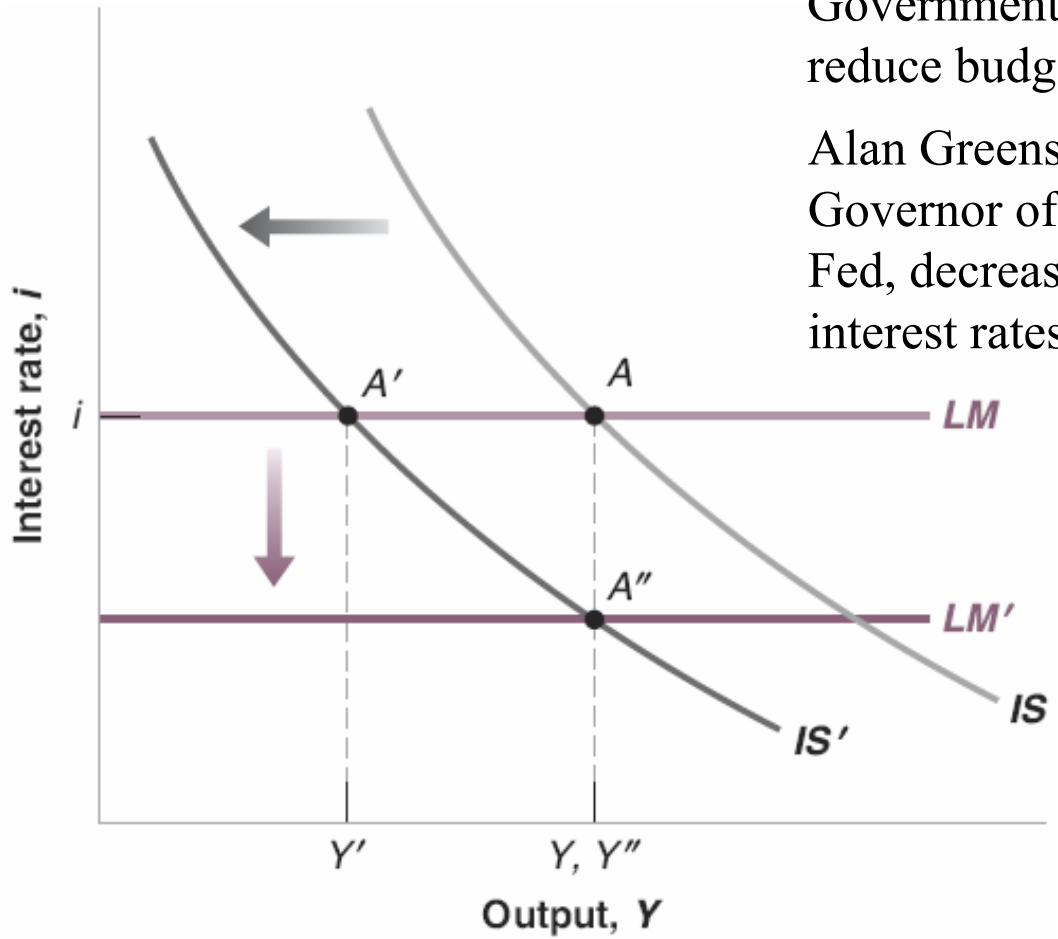
1992 Bill Clinton
Government policy:
reduce budget deficit

Alan Greenspan,
Governor of the
Fed, decrease
interest rates.

The fiscal consolidation shifts the *IS* curve to the left.

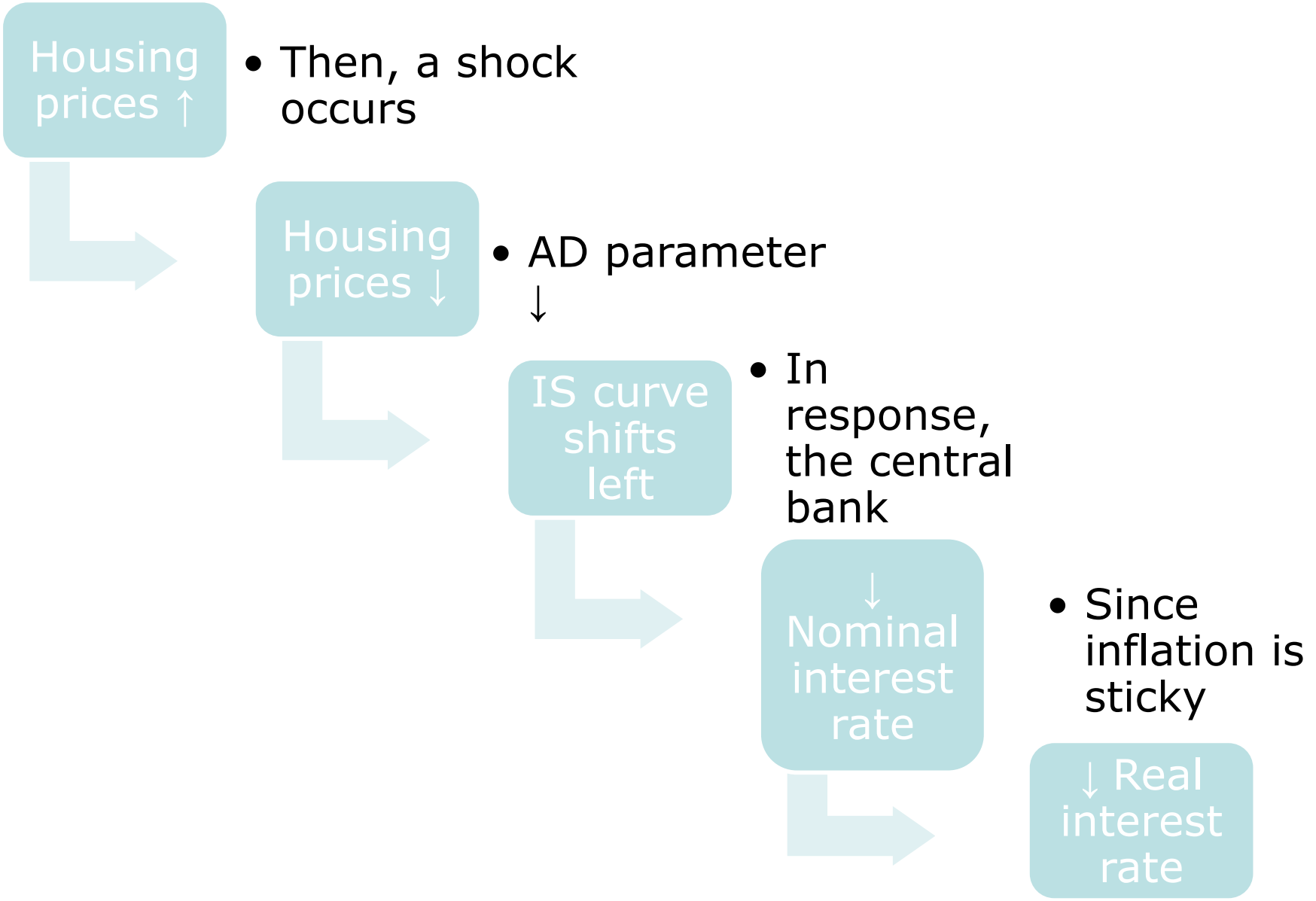
A monetary expansion shifts the *LM* curve down.

This allows for the reduction in the deficit without a recession.



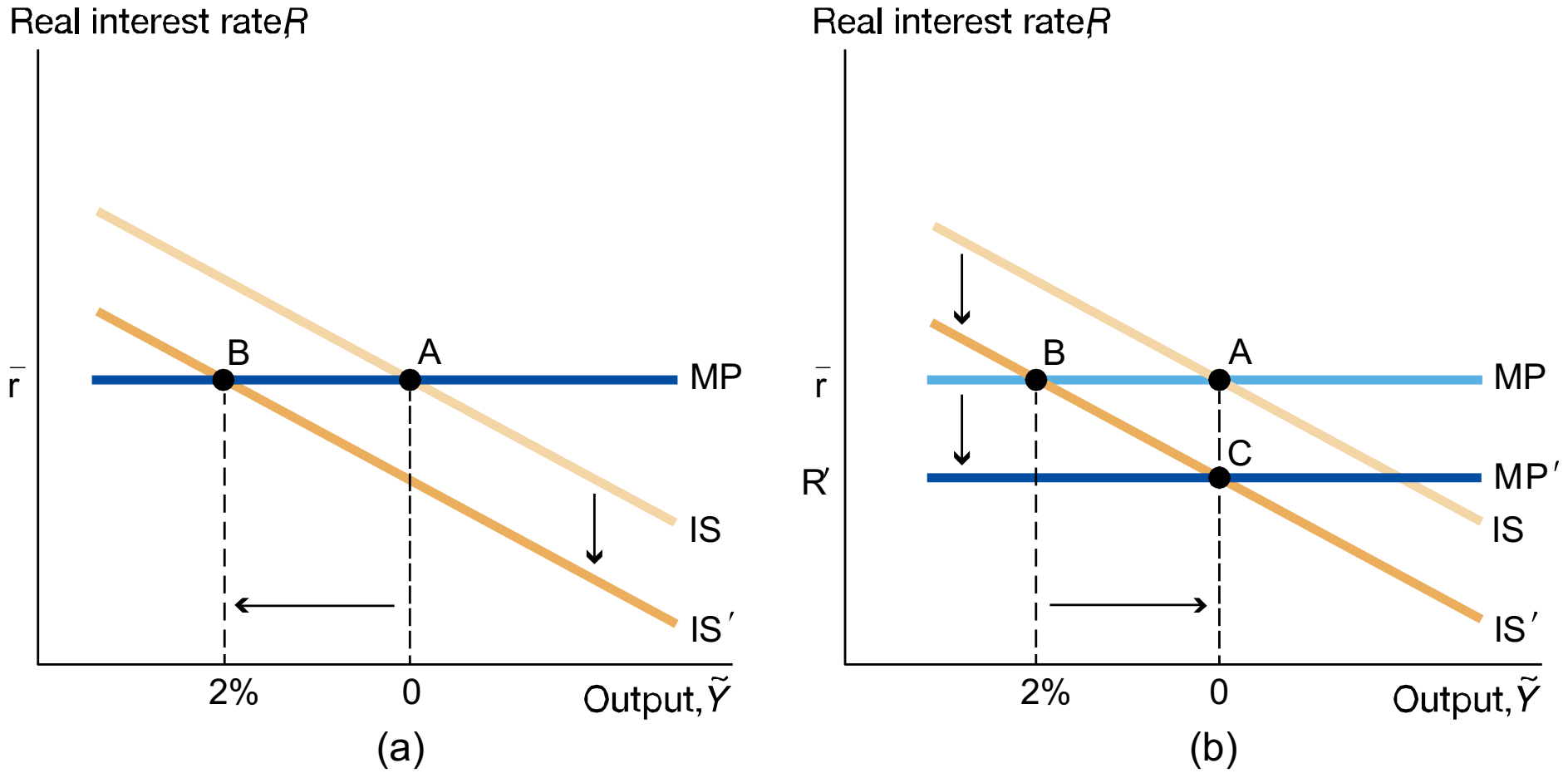
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Example: The End of a Housing Bubble



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Stabilizing the Economy after a Housing Bubble



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Exercise 1: Corona Crisis

1. Choose one country
2. Find data for the real GDP growth rate for the period 2019-2022. Show it in a table.
3. Find data for the nominal interest rate for the period 2019-2022. Show it in a table.
4. Find data for the real interest rate for the period 2019-2022. Show it in a table. If you cannot find the data, use the inflation rate and the formula: r=i-inflation rate. Show it in a table.
5. Represent the economic effects of the Lockdown in the IS-LM model.
6. Find information about macro policy response in your chosen country.
7. Represent the macro policy response in the IS-LM model.

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Exercise 2: US fiscal Package

- *One of the components of the US fiscal package was the sending of checks to households. The amount of the check was $1,200 per person, plus $500 for any dependent child. There was an income limit over which households did not qualify, but it was high, for example $150,000 for joint filers.*
- *The purpose was twofold. First, to help those households who were directly affected by the crisis, for example because one or more of the members of the households were out of work or working fewer hours. Second, to sustain aggregate demand to avoid too large a decrease in output in the non-affected sector.*

Represent the possible effect of this policy on the IS-LM model. Do you consider that it was effective?

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